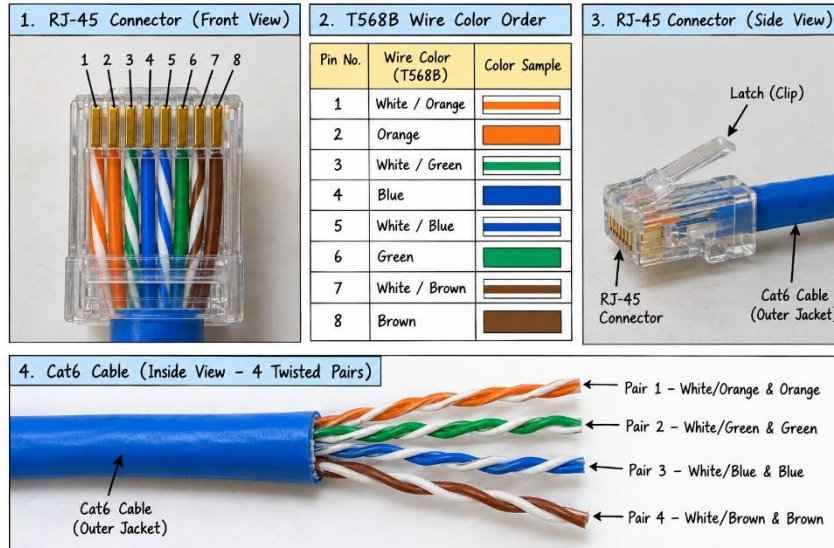


Network cable and connectors

1. Draw a labeled diagram of a Cat6 Ethernet cable, showing the color order of wires in the T568B standard and labeling the RJ-45 connector pins.



2. List three types of network cables (such as Cat5e, Cat6, and fiber optic) and for each, mention one real-world app or device (like a WiFi router, Zomato POS terminal, or IPL streaming server) where it would be used, explaining why.

1. **Cat5e Ethernet Cable:** Cat5e is commonly used to connect a **Wi-Fi router to a computer or other network device**. It is affordable, reliable, and provides enough speed for normal home and office internet use.

2. **Cat6 Ethernet Cable:** Cat6 can be used for a **Zomato POS terminal** in a restaurant. It provides faster data transfer and better performance, allowing the POS system to communicate reliably with the restaurant's network and servers.

3. **Fiber Optic Cable:** Fiber optic cables are used in **IPL streaming servers and data centers**. They support very high-speed data transfer over long distances, which is useful for delivering large amounts of video and other data with low signal loss.

3. Watch a YouTube video showing how to crimp an RJ-45 connector onto a network cable, then write down the step-by-step process in your own words and mention two common mistakes to avoid.

<https://youtu.be/QMpWkkgX1eM?si=bhF8WAYb34CZ2psV>

RJ-45 Crimping Process

1. **Cut the cable:** Cut the Cat5e/Cat6 cable to the required length, leaving some extra cable for both ends.

2. **Remove the outer jacket:** Strip about **3 cm** of the outer covering without damaging the wires inside.
3. **Prepare the wires:** Remove the plastic separator if present, untwist the four pairs, and straighten all **8 wires**.
4. **Arrange the wires:** Arrange the wires according to the **T568B standard**: White/Orange, Orange, White/Green, Blue, White/Blue, Green, White/Brown, Brown.
5. **Trim the wires:** Hold all eight wires together and cut them evenly so they are the correct length to fit inside the RJ-45 connector.
6. **Insert the RJ-45 plug:** Insert the wires into the connector while keeping the correct color order. Make sure all wires reach the end and the outer jacket goes inside the connector.
7. **Crimp the connector:** Place the RJ-45 plug into the crimping tool and press firmly. This locks the wires into the metal pins and secures the cable.
8. **Repeat and test:** Repeat the same process on the other end and use a **cable tester** to check that all 8 connections are working correctly.

Two Common Mistakes to Avoid

- **Wrong wire order:** Make sure the wires follow the T568B color order and are in the same order on both ends.
 - **Wires not inserted fully:** Ensure all 8 wires reach the end of the RJ-45 plug and that the cable jacket is inside the connector before crimping.
-
4. Given the following scenario: You are setting up a gaming zone for an IPL screening event and need to connect multiple screens and consoles. Choose the appropriate cable and connector types for high-speed, reliable connections, and explain your choices.

For an **IPL screening gaming zone**, I would use **HDMI 2.0/2.1 cables with HDMI connectors** to connect gaming consoles such as PlayStation or Xbox to the screens because they support high-quality video and audio with low delay. For the network connection between consoles, PCs, and the router/switch, I would use **Cat6 Ethernet cables with RJ-45 connectors** because they provide fast and reliable connections with low latency. If the screens or network devices are very far apart, **fiber optic cable** can be used for long-distance, high-speed connections.

5. Find and share a photo (from your home, college, or online) of any network cable or connector in use (such as behind a WiFi router or at a café's POS counter), and describe what type it is and what devices it connects.



The photo shows a **yellow Ethernet network cable with an RJ-45 connector** connected to a network router/ONT. It is used to connect the router to devices such as a **computer, switch, or another networking device**. The Ethernet cable provides a stable wired connection for transferring data over the local network.